Progress in Solving Differential Equations for Feynman Integrals

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Plan for Today

1. Introduction
2. Method of Differential Equations
   - Feynman Integrals
   - Holonomic Functions
3. Reduction to Epsilon Form
   - Lee method
   - `Fuchsia` project
4. Example
5. Summary

Introduction

Feynman Integrals Calculus — became in recent decades a science on its own.

$$\int \underbrace{\mathrm{d}^d l_1 \ldots \mathrm{d}^d l_n}_{\text{loops}} \underbrace{\mathrm{d}^d p_1 \delta(p_1^2) \ldots \mathrm{d}^d p_m \delta(p_m^2)}_{\text{legs}} \frac{1}{D_1^{n_1} \ldots D_k^{n_k}}$$

Numerical methods:

- Sector Decomposition
- Subtraction Schemes

Analytical methods

- Feynman/Schwinger/Mellin-Barnes parametrization
- Integration-By-Parts reduction Chetyrkin, Tkachov '81
  - Laporta algorithm Laporta '00: `AIR`, `FIRE`, `Reduze`
  - Symbolic reduction: `LiteRed` Lee '12
  - private implementations
- Method of Differential Equations Kotikov '91, Remiddi '97
  - Epsilon Form Henn '13
  - Lee algorithm Lee '14: `Fuchsia`, `Epsilon`
- ...

Introduction

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$\boxed{\text{Integration-By-Parts}}$ reduction

- Integral Families
  - integration momenta
    - loop – $l_1, \ldots, l_n$ only
    - phase-space – $p_1, \ldots, p_m$ only
    - mixed
  - set of denominators (topology)
  - master integrals
- Reduction
  - any integral (from the family) in terms of masters
    - $\boxed{\text{including derivatives}}$
  - completely analytical
  - highly automated

I.1 Method of Differential Equations

1. Construct System of ODE
   - from definition
   - from IBP rules
     - highly automated
     - `AIR`, `FIRE`, `LiteRed`, `Reduze2`
2. Find Epsilon Form
   - already very powerful (less-understood than IBP)
   - Lee method: `Fuchsia` (2016), `epsilon` (2017)
   - other:
3. Solve System of ODE
4. Find Constants of Integration
   - depends on the problem
   - may be the most complex task

I.2 Holonomic Functions

A function $f = f(x)$ is called *holonomic* if there exist polynomials $a_n(x), \ldots, a_0(x)$ such that

$$a_n(x) f^{(n)} - a_{n-1}(x)\, f^{(n-1)} - \ldots - a_0(x)\, f = 0$$

holds for all x. Hence, the holonomic function is uniquely defined by

- the differential equation
- a number of initial values $f(x_0), f'(x_0), \ldots, f^{(n-1)}(x_0)$

Examples of holonomic functions:

- *all algebraic functions*
- *Generalized Hypergeometric functions*
  - *polylogarythms*
  - *Elliptic functions*
- *Bessel functions*
- *Airy functions*
- *Legendre and Chebyshev polynomials*
- *Heun functions*
- and many others that have no name and no closed form

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$$ {}_{p+1}F_p \left(\begin{matrix} a_1, a_2, \ldots, a_{p+1} \\ b_1, b_2, \ldots, b_p \end{matrix} ; x \right) = \prod_{i=1}^{p} \frac{\Gamma(b_i)}{\Gamma(a_i)\Gamma(b_i - a_i)} \int_0^1 \frac{t_i^{a_i - 1}(1-t_i)^{b_i - a_i - 1}}{(1 - x\, t_1 \ldots t_p)^{a_{p+1}}} \mathrm{d}t_i $$

is a solution to the differential equation

$$\left[D\,(D + b_1 - 1) \cdots (D + b_p - 1) \;-\; x\,(D + a_1) \cdots (D + a_{p+1}) \right] y = 0$$

where

$$D = x \frac{\mathrm{d}}{\mathrm{d}x}$$

I.2 Holonomic Functions

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Conclusion

- simple representation
  - polynomials
  - ordinary differential equations
- define many complicated functions
  - no closed form
  - non-trivial integration representation
- represent Feynman integrals
- alternative for direct integration

I.2 Holonomic Functions

The equation

$$y^{(n)} - a_1(x)\, y^{(n-1)} - \ldots - a_n(x)\, y = 0$$

is easy to rewrite as a system

$$\frac{\mathrm{d}y}{\mathrm{d}x} = A(x)\, y$$

where

$$A(x) = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ a_n(x) & a_{n-1}(x) & \cdots & a_2(x) & a_1(x) \end{bmatrix} \quad \text{and} \quad y = \begin{pmatrix} y \\ y' \\ \vdots \\ y^{(n-2)} \\ y^{(n-1)} \end{pmatrix}$$

II.1 Epsilon Form

- Classical Notation

$$\frac{\mathrm{d}F_1}{\mathrm{d}x} = A_{11}(x,\epsilon)F_1 + A_{12}(x,\epsilon)F_2$$
$$\frac{\mathrm{d}F_2}{\mathrm{d}x} = A_{21}(x,\epsilon)F_1 + A_{22}(x,\epsilon)F_2$$

- Matrix Notation

$$\frac{\mathrm{d}\bar{F}}{\mathrm{d}x} = A(x,\epsilon)\,\bar{F} \qquad \text{where} \qquad A = \begin{pmatrix} A_{11}(x,\epsilon) & A_{12}(x,\epsilon) \\ A_{21}(x,\epsilon) & A_{22}(x,\epsilon) \end{pmatrix} \quad \text{and} \quad \bar{F} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$$

It is very convenient (as observed by J.Henn) to have our system in the epsilon form

$$\frac{\mathrm{d}G}{\mathrm{d}x} = \epsilon\, B(x)\, G$$

since in this case we can easily find the solution to any order in ϵ parameter, as we will see on the next slide.

Some physical examples may lead to systems with ~ 500 equations. Hence, it is very important to make this task automatic.

II.2 Solutions

We are looking for the solution of a given system of ordinary differential equations in the epsilon form

$$\frac{\mathrm{d}G}{\mathrm{d}x} = \epsilon\, B(x)\, G$$

as a Laurent series in ϵ

$$\boxed{G(x,\epsilon) = G_0(x) + G_1(x)\,\epsilon + G_2(x)\,\epsilon^2 + \ldots}$$

Let us put this "solution" into the initial equation

$$\frac{\mathrm{d}G_0}{\mathrm{d}x} + \frac{\mathrm{d}G_1}{\mathrm{d}x}\epsilon + \frac{\mathrm{d}G_2}{\mathrm{d}x}\epsilon^2 + \ldots = \epsilon\, B(x)\, G_0 + \epsilon^2\, B(x)\, G_1$$

we get

$$\frac{\mathrm{d}G_0}{\mathrm{d}x} = 0, \qquad \frac{\mathrm{d}G_1}{\mathrm{d}x} = B(x)G_0, \qquad \frac{\mathrm{d}G_2}{\mathrm{d}x} = B(x)G_1 \qquad \ldots \qquad \frac{\mathrm{d}G_n}{\mathrm{d}x} = B(x)G_{n-1}$$

This system can be easily solved (as promised)

$$G_0 = C_0, \qquad G_1 = C_1 + \int \mathrm{d}x B(x) C_0, \qquad G_2 = C_2 + \int \mathrm{d}x B(x) \left(C_1 + \int \mathrm{d}x B(x) C_0 \right) \qquad \ldots$$

$$\boxed{G_n(x) = C_n + \int \mathrm{d}x B(x) G_{n-1}}$$

II.3 Project `Fuchsia`

Input

- System of Ordinary Differential Equations $A(x,\epsilon,\ldots)$, i.e.,

$$\frac{\mathrm{d}F}{\mathrm{d}x} = A(x,\epsilon,\ldots)\,F(x,\epsilon,\ldots)$$

Output

- Equivalent System in the Epsilon Form

$$\frac{\mathrm{d}G}{\mathrm{d}x} = \epsilon\,B(x,\ldots)\,G(x,\epsilon,\ldots)$$

- Corresponding Basis Transformation

$$F(x,\epsilon,\ldots) = T(x,\epsilon,\ldots)\times G(x,\epsilon,\ldots)$$

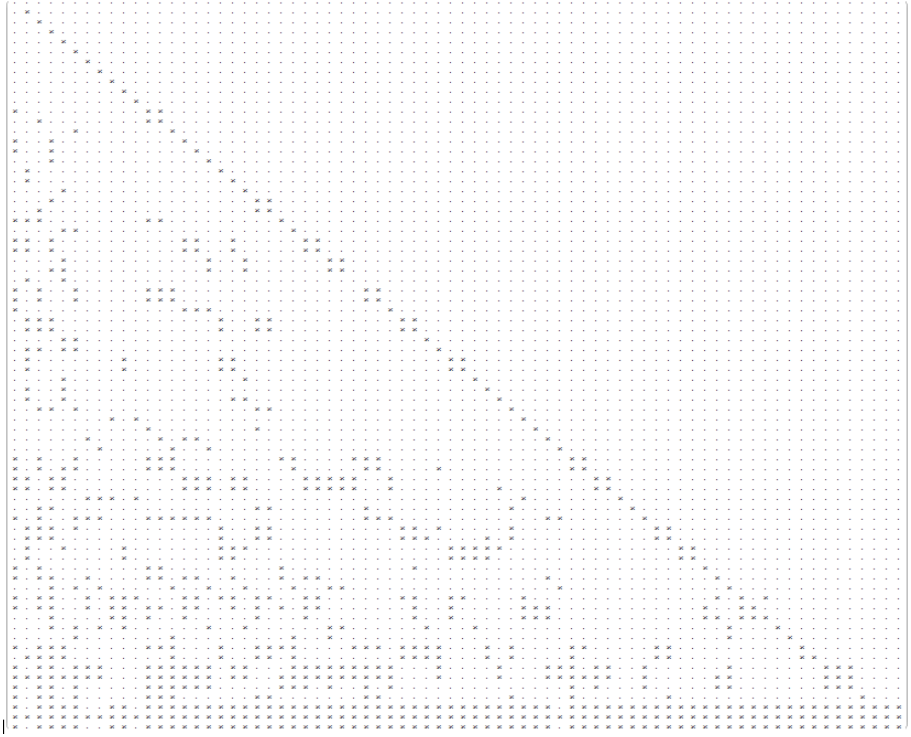
- Other Operations
  - apply custom transformation
  - variable change
  - "sort" to block-diagonal form

II.3 Project `Fuchsia`

- Based on the *Lee algorithm* Lee '14
  - support additional symbols
  - alternative implementation: `epsilon`
- Open-Source and Free Gituliar, Magerya '16 '17
  - `http://github.com/gituliar/fuchsia`
- Implemented in `Python`
  - `SageMath`
  - `Maxima`
  - `Maple` (optional)
- Algorithm
  1. **Fuchsification** (Jordan form)
     Get rid of apparent singularities
  2. **Normalization** (eigenvalues, eigenvectors)
     Balance eigenvalues to $\alpha\,\epsilon$ form
  3. **Factorization** (solve linear equations)
     Reduce to the epsilon form

II.3 Project Fuchsia

Non-planar double pentabox diagram Papadopoulos, Tommasini, Wever ’16



Step I: Fuchsification

We say that a given matrix has *Fuchsian form* if its Poincaré rank is 0 at every singular point (including ∞).

Fuchsification is based on analysis of the Jordan form of a given matrix.

For example, we can transform

$$\begin{pmatrix} \frac{\epsilon}{x} & 0 \\ -\frac{\epsilon}{x^2} & \frac{\epsilon}{1+x} \end{pmatrix} \to \begin{pmatrix} \frac{\epsilon}{x} & 0 \\ -\frac{1}{x} & \frac{\epsilon}{1+x} + \frac{1}{x} \end{pmatrix}$$

which contains only regular singularities.

Unfortunately, not every system can be transformed to the Fuchsian form

- due to the analyticity of S-matrix; and
- structure of the Feynman integrals

in principle, every system for Feynman integrals should be reducible to Fuchsian form.

In practice things are more complicated...

Step II: Normalization

We say that matrix $\boldsymbol{A(x,\epsilon)}$ **is normalized** if eigenvalues of all its residues have form $m\epsilon$, where m is some number.

We assume that initial eigenvalues have form $n + m\epsilon$, where $\boldsymbol{n}$ **is integer**

A key idea for Lee's normalization algorithm is a **balance transformation** between two points x_1 and x_2

$$T(x) = B(P, x_1, x_2; x) = 1 - P(\epsilon) + \frac{x - x_2}{x - x_1} P(\epsilon)$$

where $P(\epsilon)$ is some projector matrix, i.e. $P^2 = P$.

We choose

- points x_1 and x_2 by analyzing eigenvalues
- projector $P(\epsilon)$ by analyzing eigenvectors

Example of eigenvalues in different points

$x_1 = 0$:

$[1-4\epsilon,\ 1-3\epsilon,\ -1-2\epsilon,\ -2\epsilon,\ 1-2\epsilon,\ 1-2\epsilon,\ 1,\ 1]$

$x_2 = 1$:

$[-1-2\epsilon,\ -2\epsilon,\ -2\epsilon,\ 1-2\epsilon,\ 1-2\epsilon,\ 0,\ 0,\ 0]$

Step III: Factorization

Now we can find an x-independent transformation $T(\epsilon)$ for any point $x = x_k$ such that

$$T^{-1}(\epsilon)A_k(\epsilon)T(\epsilon) = \epsilon C_k$$

Since matrix C_k is constant, for every residue of A we write

$$\frac{T^{-1}(\epsilon)A_k(\epsilon)T(\epsilon)}{\epsilon} = C_k = \frac{T^{-1}(\mu)A_k(\mu)T(\mu)}{\mu}, \quad \mu = \text{any number}$$

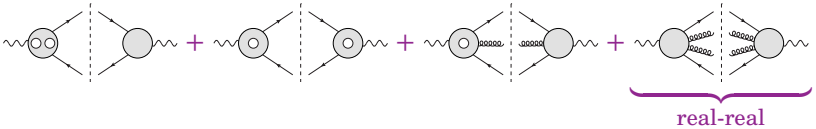
We treat components of T as unknown variables and solve linear system of equations for them. That gives unknown transformation $T(\epsilon)$.

Example I: Splitting Functions at NLO in QCD

Splitting Functions for DGLAP evolution equations can be extracted from the

$$e^+e^- \to \gamma^* \to \text{partons}$$

annihilation process:



Initial Form

$$\begin{pmatrix}
\frac{(2\epsilon-1)(2x-1)}{x(1-x)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\epsilon-2}{x(1-x)} & \frac{1-2\epsilon}{x} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{(2\epsilon-1)}{\epsilon^2 x(1-x)(x+1)} & 0 & \frac{1-6\epsilon}{x+1} & \frac{2}{x+1} & 0 & 0 & 0 & 0 \\
\frac{(2\epsilon-1)}{\epsilon x^2(x+1)} & \frac{(1-2\epsilon)(3\epsilon-1)}{x^2} & \frac{2(6\epsilon-1)}{x(x+1)} & \frac{2\epsilon(x^2+3x-2)}{(1-x)x(x+1)} & 0 & 0 & 0 & 0 \\
\frac{2(x^2+4x+1)}{\epsilon^2(1-x)x^3(x+1)^3} & \frac{2(2\epsilon-1)(x-1)}{\epsilon x^3(x+1)^2} & \frac{2(6\epsilon-1)(x-1)}{x^2(x+1)^3} & \frac{4(x^2+1)}{x^2(x+1)^3} & -\frac{(2\epsilon+1)(2x+1)}{x(x+1)} & 0 & 0 & 0 \\
0 & -\frac{(2\epsilon-1)}{\epsilon(1-x)x} & 0 & 0 & 0 & \frac{2\epsilon}{1-x} & 0 & 0 \\
\frac{-4}{\epsilon^2(1-x)^3x^3(x+1)} & \frac{2(1-2\epsilon)(x-2)}{\epsilon(1-x)^2x^3} & \frac{2(1-6\epsilon)}{x^2(1-x)(x+1)} & \frac{4(x^2+1)}{(1-x)^2x^2(x+1)} & 0 & \frac{-4\epsilon}{(1-x)^2x} & \frac{(2\epsilon+1)(2x-1)}{(1-x)x} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{(2\epsilon+1)(2x-1)}{(1-x)x}
\end{pmatrix}$$

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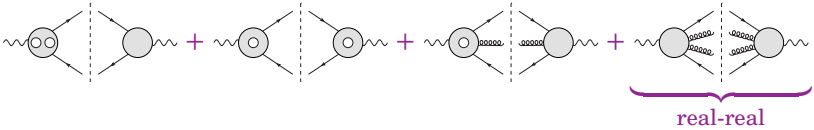

Epsilon Form

$$\begin{pmatrix}
\frac{2}{1-x}-\frac{2}{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{1}{19x} & -\frac{3}{x} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{1}{19(1-x)} & \frac{2}{1-x} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2}{1-x}-\frac{2}{x} & 0 & 0 & 0 & 0 \\
\frac{3}{x} & -\frac{19}{x} & 0 & 0 & \frac{2}{1-x}+\frac{4}{x} & -\frac{1}{x} & 0 & 0 \\
\frac{6}{1+x}+\frac{6}{1-x}+\frac{16}{x} & \frac{76}{1+x}-\frac{152}{x} & 0 & 0 & \frac{12}{1+x}+\frac{12}{1-x}+\frac{32}{x} & -\frac{2}{1+x}-\frac{8}{x} & 0 & 0 \\
\frac{2}{19(1-x)}-\frac{4}{19x} & \frac{4}{(1-x)}+\frac{8}{x} & \frac{76}{x}-\frac{76}{1-x} & \frac{4}{19(1-x)}-\frac{16}{19x} & 0 & \frac{3}{19x} & \frac{2}{1-x}-\frac{2}{x} & 0 \\
\frac{8}{19x}-\frac{6}{19(1+x)} & \frac{4}{1+x}-\frac{4}{x} & 0 & 0 & \frac{4}{19(1-x)}+\frac{16}{19x}-\frac{8}{19(1+x)} & \frac{2}{19(1+x)}-\frac{3}{19x} & 0 & -\frac{2}{1+x}-\frac{2}{x}
\end{pmatrix}$$

We can easily solve this system and find master integrals OG '16.

Summary

`Fuchsia` — a tool for reducing differential equations for master integrals to the epsilon form Henn '13:

- complete implementation of Lee '14 algorithm
- open-source and free (`Python`, `SageMath`, `Maxima`)
- powerful, e.g., two-loop planar pentabox
- `http://github.com/gituliar/fuchsia`
- `arXiv:1607.0079`, `arXiv:1701.04269`

Questions?